

Racketlon Match Prediction

Dynamic Ratings, Machine Learning Models, and Confidence Analysis

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Problem

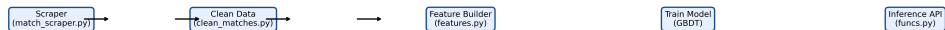
Why is this hard?

Racketlon combines four sports—table tennis, badminton, squash, and tennis—into a single total-points competition.

- Player skill is inherently **multi-dimensional**
- Match outcome depends on both **sport-level strengths** and **aggregate scoring**
- Performance changes over time, so static skill estimates are not enough

Goal: predict sport-level score differences, total match score difference, and the final winner.

Pipeline



- scrape historical match data
- clean and normalize records
- build chronological leakage-safe features
- train multiple models
- evaluate accuracy and confidence

Data and Features

Data pipeline

- scraped Tournament Software pages
- handled both modern and legacy formats
- removed invalid / incomplete matches
- normalized player identities

Feature groups

- sport-specific rating differences
- recent rolling form
- residual performance
- long-term averages
- head-to-head features

Important design choice

All features were built **chronologically**, so each match only uses information available before it occurred.

Dynamic Rating System

For each sport, each player has a separate rating.

Expected score difference:

$$\hat{d} = 21 \tanh \left(\frac{\alpha(R_1 - R_2)}{2} \right)$$

Experience-dependent learning rate:

$$\eta(g) = \eta_{\min} + (\eta_{\max} - \eta_{\min})e^{-g/\tau}$$

Time-based adjustment:

$$t_{\text{mult}} = 1 + c_s \left(1 - e^{-d/\tau_s} \right)$$

- gives interpretable sport-specific skill estimates
- adapts to both experience and inactivity
- provides strong predictive features for downstream models

Models Evaluated

Baseline

- historical average score differences

Linear model

- ridge regression on engineered features

Player embedding model

- learned latent player representations

Boosted tree model

- final and strongest model
- base prediction + residual correction

Confidence experiment

- KD-tree neighborhood analysis
- tested whether local density predicts reliability

$$\hat{d} = b + 0.7r$$

for the final boosted tree differential prediction

Main Results

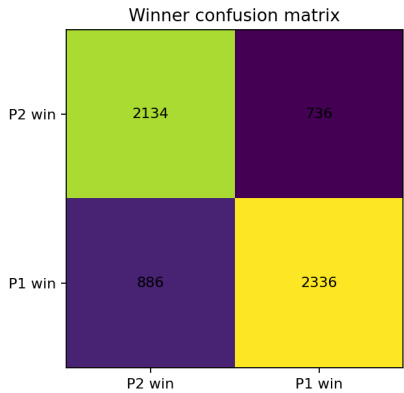
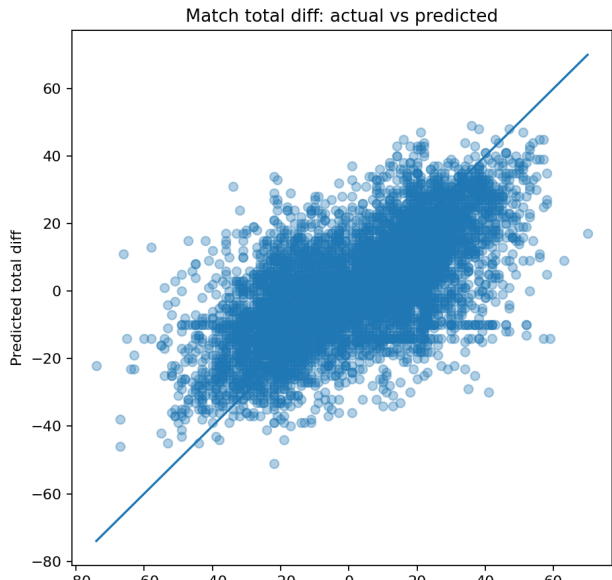
Best model: gradient boosted trees

- match-level MAE: **8.27**
- winner accuracy: **89%**
- baseline was weakest
- linear and embedding models improved performance
- boosted trees performed best by capturing nonlinear interactions

Takeaway

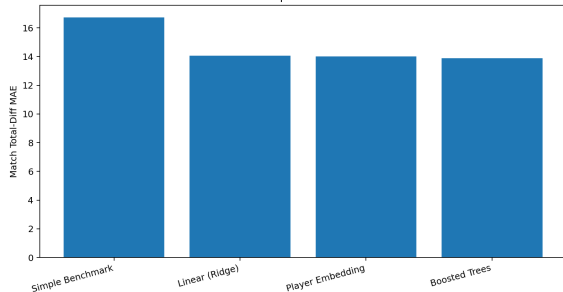
Combining a strong rating system with nonlinear machine learning gave the best predictive performance.

Prediction Quality

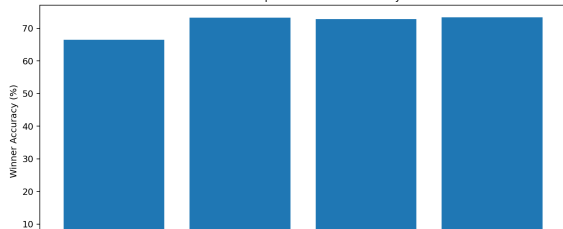


Model Comparison and Interpretability

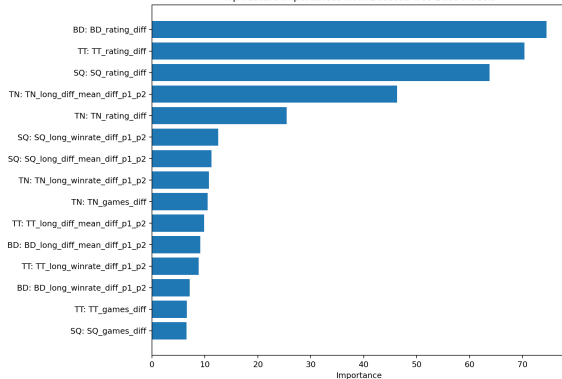
Model Comparison: Match MAE



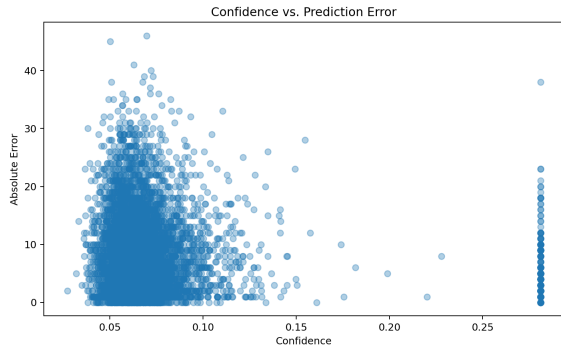
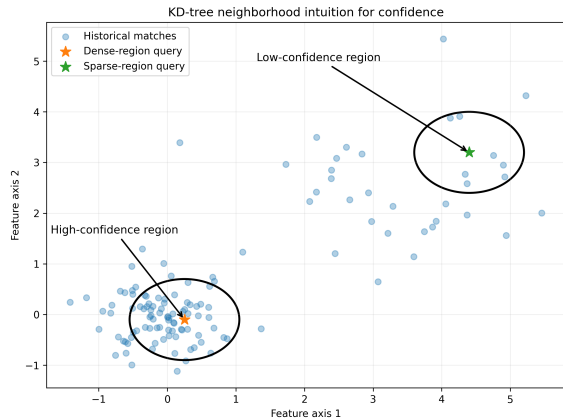
Model Comparison: Winner Accuracy



Top Feature Importances from Boosted Tree Base Models



Exploratory Confidence Experiment



Question

Do predictions made in dense regions of feature space have lower error?

Conclusion

- Built a full end-to-end Racketlon prediction pipeline
- Dynamic sport-specific ratings provided a strong foundation
- Gradient boosted trees gave the best overall performance
- Simple KD-tree confidence measures were not very effective for the final model

Main lesson

The best performance came from combining domain-specific structure with a flexible nonlinear model.

Questions?